Linear Programming

Design and Analysis of Algorithms

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Definition

- Optimization of a linear objective function, subject to linear equality and inequality constraints.
- Real-world examples (e.g., resource allocation, scheduling, diet problem).
- Standard form:

$$\text{Maximize (or Minimize) } c^T x$$

$$\text{subject to } Ax \leq b, x \geq 0$$

$$x_i \in \mathbb{R} \text{ for } i \in \{1, 2, \ldots, n\}$$

  - $x = x_1, x_2, \ldots, x_n$ are the decision variables
  - c is the coefficient vector for the objective function
  - A and b define the linear constraints

Example

- Choose a combination of foods to minimize cost, while ensuring minimum nutritional requirements (like calories, protein, vitamins, etc.) are met.
- Let:
  - n: number of available foods, m: number of nutrients (constraints)
  - x_i : quantity of food item i to include in the diet (decision variables)
  - c_i: cost per unit of food item i
  - a_{ji}: amount of nutrient j in one unit of food item i
  - b_j : required amount of nutrient j
- Linear Program:

$$\text{Minimize } Z = \sum_{i=1}^{n} c_i x_i$$

subject to (nutrient constraints):

$$\sum_{i=1}^{n} a_{ji} x_i \geq b_j \text{ for } j = 1, 2, \ldots, m$$

$$x_i \geq 0 \text{ for } i = 1, 2, \ldots, n$$

Example (cont.)

- Suppose we have two foods A and B as shown below:

Food	Cost	Protein (g)	Calcium (mg)
Food A	50	3	2
Food B	20	2	4
Min Req		8	6

- Let x_1 and x_2 be the units of Food A and B, resp, to be consumed. Then the linear program would be:

$$\text{Minimize } Z = 50x_1 + 20x_2$$

subject to:

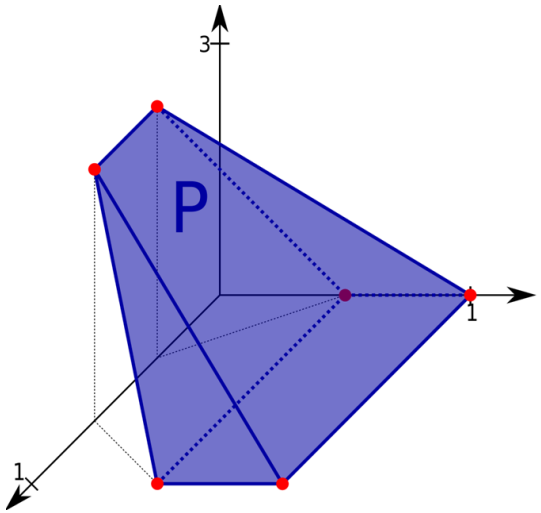
$$3x_1 + 2x_2 \geq 8 \text{ (protein)}$$

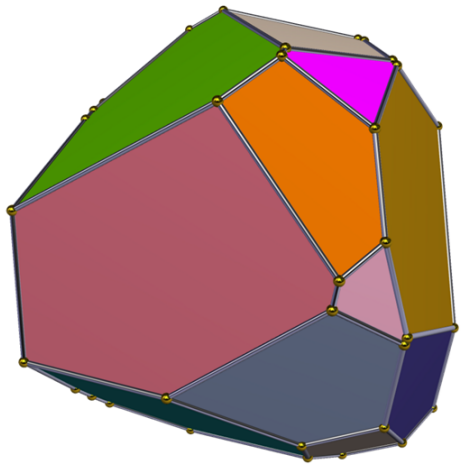
$$2x_1 + 4x_2 \geq 6 \text{ (calcium)}$$

$$x_1, x_2 \geq 0$$

Solving Linear Programs - Simplex method

- The feasible region of an LP is a convex polytope.

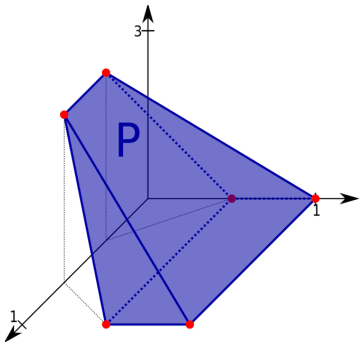




- The maximum (or minimum) value of a linear objective function over this region must occur at a vertex (corner point).
- So, instead of checking every point, we just look at the corners.
- The simplex method moves from one vertex to an adjacent vertex with better objective value.
- Imagine standing inside a multi-sided shape (a polyhedron) in some dimensional space.
- Each face of that shape represents a constraint (like $x + y \leq 5$), and your goal

Solving Linear Programs - Simplex method (cont.)

- Imagine standing inside a multi-sided shape (a polyhedron) in some dimensional space.



- Each face of that shape represents a constraint (like $x + y \leq 5$), and the goal is to climb to the highest possible value of some function (like profit or efficiency), which is the objective function.
- The Simplex Method moves from vertex to vertex, choosing the direction that most improves the objective.
- It's like hiking up a mountain made of flat faces, and always choosing the neighboring peak that gives you the steepest increase in elevation — until you can't go any higher.

Why is Simplex Method smart?

- It doesn't search blindly. It's very strategic — only considers directions that can possibly improve the objective.
- It avoids unnecessary points — instead of checking all corners, it uses gradients (objective function coefficients) to guide its path.
- It's efficient in practice, even if worst-case complexity is exponential.

Integer Linear Program

- An Integer Linear Program (ILP) is a type of optimization problem where:
  - The objective function is linear
  - The constraints are linear equalities or inequalities
  - Some or all of the decision variables must take integer values
- Standard form:

$$\text{Maximize (or Minimize) } c^T x$$

$$\text{subject to } Ax \leq b, x \geq 0$$

$$x \in \mathbb{Z}^n$$

  - $x = x_1, x_2, \ldots, x_n$ are the decision variables
  - c is the coefficient vector for the objective function
  - A and b define the linear constraints
  - $\mathbb{Z}^n$ = integer space (all variables must be integers)

LP vs ILP

Feature	Linear Programming (LP)	Integer Linear Programming (ILP)
Variables	Can take any real values (fractions, decimals, etc.)	Must take **integer** values (can also be binary or general)
Solution Space	Continuous and convex polyhedron	Discrete points inside the polyhedron
Feasibility Region	Convex, with infinitely many points	Non-convex, with finitely many feasible integer points
Optimal Solution	Found at a **vertex** (corner point) of the feasible region	May not lie at the vertex of LP relaxation
Solving Methods	Simplex, Interior Point Methods	Branch & Bound, Cutting Planes, Branch & Cut
Computation Time (Generally)	Polynomial-time (practical cases)	NP-hard (exponential in worst case)
Ease of Solving	Relatively easy and fast	Much harder and more computationally intensive
Applications	Resource allocation, transportation, blending	Scheduling, assignment, routing, knapsack-type problems

Summary

- Definition of a Linear Program
- Example
- Simplex Method
- Integer Linear Program
- LP vs ILP